

**STRULE AUTOMATION**

Industrial Controls · Bay Area &amp; Northern California

## DIAGNOSTIC FIELD GUIDE

# Tracking Down the Fault That Won't Repeat

**Who this is for and what it does:** a maintenance tech, controls engineer, or plant reliability person who keeps getting handed a fault that clears itself before anyone reaches the panel. This walks you through capturing evidence, narrowing the cause, and setting a trap so the next occurrence actually tells you something.

## How to use this

Read the triage order once. When the fault hits, work from evidence, not the last thing someone swapped. If it's already cleared, jump to "Set a trap" and get instrumentation in place first.

The frustrating thing about an intermittent is that the failure and the diagnosis rarely coincide. By the time you reach the panel with a laptop, the terminal is tight again, the condensation has burned off, the noise source has cycled off, and everything meters fine. So the first job is never "fix it." The first job is to make the fault leave a record.

## Step one: capture evidence before you touch anything

An intermittent won't show itself on demand, so before you start swapping parts, get these working:

- **First-out / first-fault logging.** Most controllers and safety systems record which condition tripped first in a cascade. One device faulting looks identical to ten until first-out tells you which moved first. Turn it on if it isn't already; if your platform is an older or basic one that has no first-out at all, note that and lean on the external trap described at the end of this guide.
- **Event and alarm logging with real timestamps.** Confirm the clock is right and, if you have more than one device, that they're synchronized. A programmable logic controller (PLC) log that says 14:32:07 and a drive log that says 14:29:51 for the same event will send you chasing a phantom. Fix time sync before you trust any correlation.
- **Trending on the suspect signals.** Log the analog values, comms status bits, supply voltages, and digital inputs around the fault. Sample fast enough to catch the event you're chasing: a once-per-second trend misses a 200-millisecond dropout entirely, and even a 200-ms sample can straddle and miss it. To reliably catch a glitch that short, sample well under 100 ms, or better, use a

triggered or edge-latched capture that freezes on the event instead of hoping a periodic sample lands on it.

- **The human log.** Ask the operators the boring questions and write the answers down. Time of day? Startup or steady running? Wet weather? Big compressor kicking in? They often know the pattern already and nobody wrote it down.

Timestamps are the whole game. Tie the fault to a moment, then find something else that happened at that same moment, and you've turned a mystery into a lead. I once burned two visits on a dropout that turned out to track a chiller in the next room cycling on; the operator had clocked the timing weeks earlier, I just hadn't asked.

## The usual families of cause

Intermittents cluster into a handful of recurring stories. Here's how they present:

| Symptom you see                                                                                                        | Likely causes                                                                                                  | What to check / what to log                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fault correlates with vibration, machine motion, or a passing forklift; comes and goes when nothing electrical changed | Loose termination, cracked solder joint, chafed conductor, a spade lug that's backing off                      | Tug-test terminations (power off). Thermal imaging only finds a loose joint on a current-carrying power termination under real load — a motor feed, a supply rail, a contactor lug; on a milliamp signal or a lightly-loaded 24 VDC digital-input terminal a bad joint dissipates almost nothing and images stone cold, which reads as a false all-clear. On those signal-level circuits, measure the millivolt drop across the joint itself, or wiggle-test with a trend running and log the digital input state. |
| Fault appears in early morning, after a cold night, or in humid weather; clears once things warm up                    | Condensation inside an enclosure, moisture on a terminal block, a failed heater, connector corrosion           | Check the enclosure heater and thermostat. Look for water tracking, green corrosion, a failed door gasket. Log enclosure temperature and humidity.                                                                                                                                                                                                                                                                                                                                                                 |
| Fault tracks with temperature, worse on hot afternoons or near a heat source                                           | Marginal component drifting with temperature, an undersized or failing supply sagging when warm, a stopped fan | Log cabinet and ambient temperature against the fault. Check supply rails under load when hot. Confirm cooling fans and filters are moving air.                                                                                                                                                                                                                                                                                                                                                                    |

| Symptom you see                                                                                                     | Likely causes                                                                                                                                                                  | What to check / what to log                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analog reading noisy or jumpy; false trips with no real process change; problem started after new equipment went in | Ground loop, missing or double-ended shield, a variable frequency drive (VFD) injecting noise, poor bonding                                                                    | Verify shields are landed at one end only, per the device's convention. Trace the bonding path to a single point. Meter for stray millivolts between two grounds that should match; a difference points at a ground loop. Log the raw analog value, not the scaled one. |
| A device drops off the network, then comes back; comms-loss alarm that self-clears; whole segment stutters          | Marginal cable or connector, a flaky media converter or switch port, a duplicate IP address, wrong termination/biasing on a serial or fieldbus link, cable near a noise source | Log comms health bits and error counters. Check switch port stats for CRC errors and flapping. On Ethernet, scan for a duplicate address. Inspect connectors and strain relief. Confirm bus termination resistors are correct.                                          |
| A single measurement slowly wanders, or reads wrong only under certain process conditions                           | Sensor drifting, fouling, or coating; a thermocouple or RTD connection going high-resistance; a transmitter near its range limit                                               | Trend the raw signal over days, not minutes. Compare against a second instrument or a manual reading. Pull the sensor and inspect for buildup or moisture in the head.                                                                                                  |
| Multiple unrelated devices glitch or reset at the same instant; resets nobody commanded                             | Power dip or sag, brownout on a shared supply, a failing 24 VDC supply, inrush from a large load starting                                                                      | Log the 24 VDC and control-supply voltage fast. Look for what large load starts at that moment. Check for a supply shared past its rating. A power-quality logger on the incoming feed earns its keep here.                                                             |

RTD above is a resistance temperature detector. CRC is a cyclic redundancy check, the network's built-in error count; rising CRC errors mean the cable or connector is marginal. Two devices set to the same IP (internet protocol) address will each drop off as they fight for it, which reads just like a flaky cable.

## A practical triage order

At the panel, work from cheapest and most common toward rarest and most invasive:

1. **Read the logs first.** First-out, alarm history, drive fault buffers, event logs. Get the timestamp before you get out a screwdriver.
2. **Look and smell.** Open the enclosure. Water tracks, corrosion, a burnt smell, a heat-discolored terminal, a wire hanging by two strands. A lot of intermittents are visible if you actually look.
3. **Check power and grounding.** Meter the control supplies under load. Confirm the bonding path. I once watched an analog input flap for weeks after a VFD went in near the tray; it came down to a

shield landed at both ends. These underlie a surprising share of "weird" faults and are quick to rule out.

4. **Check connections mechanically.** Tug-test, retorque to spec (power off), and thermal-image under load — but only where the joint carries real current; a milliamp signal or a lightly-loaded DI terminal images cold even with a bad joint, so fall back to a millivolt drop across the joint or a wiggle-test-with-a-trend there. The longest intermittent I ever chased was one spade lug backed off a quarter turn; it only dropped out once the panel warmed up, and imaged warm the second I put load on it.
5. **Check comms.** Error counters, port stats, cable and connector condition, termination.
6. **Check the sensor / final element last.** Instruments get blamed first and are usually innocent. Prove it with a second reading before you condemn one.

If a step turns up something concrete, stop and pursue it. Treat the order as a default. If the operators tell you it only happens in the rain, start with condensation and grounding.

## Set a trap for the next occurrence

If it's already cleared, this is the real work: build something that's recording when the fault comes back.

- **Trend the right signals fast enough.** Data-log the suspect analogs, supply voltages, comms heartbeats, and digital inputs around the fault. Sample well under 100 milliseconds if you're chasing a ~200 ms glitch — sampling in the same range as the event aliases and misses it — or better, arm a triggered capture (below) so the event latches itself instead of relying on a periodic sample landing on it.
- **Latch the first-out.** Make sure whatever tripped first is held until someone reads and clears it. A self-clearing fault that leaves no first-out record is a wasted occurrence.
- **Add a trigger.** If the controller supports it, set a triggered capture on the fault bit or an out-of-range value; it grabs a data window bracketing the moment automatically.
- **No first-out, event log, or triggered capture on this PLC? Bring your own.** A large share of the installed base is older or basic and has none of the above, and that doesn't leave you stuck — a standalone datalogger, a bench scope or meter with min/max and a triggered mode on the suspect signal, a plug-in power-quality logger on the supply, or a spare fast input mirrored to a chart will each catch what the controller can't. The goal is a timestamped record around the moment, wherever it comes from.
- **Instrument the physical suspects.** A cheap temperature/humidity logger in the enclosure. A power-quality logger on the feed if you suspect dips. A clamp meter with min/max hold on a supply. These run unattended and catch what a laptop that's gone home can't.
- **Write down what "normal" looks like now.** A baseline trend from a good day makes the bad-day trend readable. Without it you're guessing at what changed.

Then it's a timestamp, a first-out, and a trend around the moment: an ordinary fault you can chase.

## 10-minute triage checklist

Run this before you call anyone:

- ☐ Note the exact time the fault last occurred and what the machine was doing.
- ☐ Pull first-out / first-fault and the alarm/event log; screenshot or export it.
- ☐ Confirm device clocks are set and roughly synchronized.
- ☐ Open the enclosure and look for water, corrosion, heat marks, loose or discolored terminals.
- ☐ Meter the 24 VDC and control supplies under load; note anything sagging.
- ☐ Check comms error counters / switch port stats for the affected devices.
- ☐ Ask two operators for the pattern (time of day, weather, nearby equipment starting).
- ☐ If it can't be reproduced now, set a fast trend + first-out latch before you leave.

## Safety note

The intermittent that keeps nuisance-tripping a protective function is exactly the one someone will be tempted to jumper out just to stop the calls. If a trip is genuinely spurious and the process never approached the protected condition, retuning that threshold or filtering that input is legitimate engineering, done deliberately and documented. Forcing a safety input, bypassing a safety relay, or strapping out an interlock to silence the alarm removes the protection while you still don't know why it fired. If a safety function keeps tripping, treat each trip as evidence and find out what it's actually seeing.

## Honest limits

This guide narrows the search but won't name your specific fault — some intermittents, like a hairline crack that only opens at one temperature, take several logged occurrences to corner even with good instrumentation, and nothing here replaces your equipment's manuals, your site's safety procedures, or a qualified person's judgment on your installation.

## If it keeps getting handed around

The worst version bounces between the controls vendor, the drive vendor and the instrument supplier — each tests their own box, finds it fine, and hands it back — while the fault lives in the seams between them. That's where a vendor-neutral look helps. If you're stuck in that loop, reach me at [struleautomation.com](mailto:info@struleautomation.com) or [info@struleautomation.com](mailto:info@struleautomation.com).

— Jonathan Gilmour, Strule Automation